8|1|24

Derivatives

Derives

Forward contract

options contract

Sell / Buy the underlying
T → ?
How much, Price

can be commodity, stock etc.

0 T (Maturity date)

Forward contract

↳ one on one

Futures contract → standardised, so can be traded on exchange like NSE

Interest rate derivatives → on bonds

Buyer of futures → Bullish
Seller " " → Bearish

So "zero sum games"

Near month → expire in 1 month
Mid month → " " 2 month
Far month → " " 3 month

only valid for futures contracts which are traded on exchanges

OTC → can even have 6 months or more

Lot size → no. of underlying units which will be bought / sold

↳ Ex If you buy a contract with a lot size 50, then through one contract, you are buying, 50 units of NIFTY.

Nifty → ₹ 21,513 (Spot price, U/L price)

On $25\frac{1}{4}$, buy/sell Nifty @ 21,569.95 (futures price)

$25\frac{1}{24}$ — 21,580 → gain for buy side $S_T - K$ → contract price

↳ spot price at that time

— 21,530 → gain for sell side → $K - S_T$

↳ loss exactly same as gain

We get rid of volatility, by buying futures
So, futures can have -ve/+ve value

10/1/24

Position limits → a cap of number of future contracts a person can buy both on buy or sell side.
Price limits

₹ 6207/BBL (K)

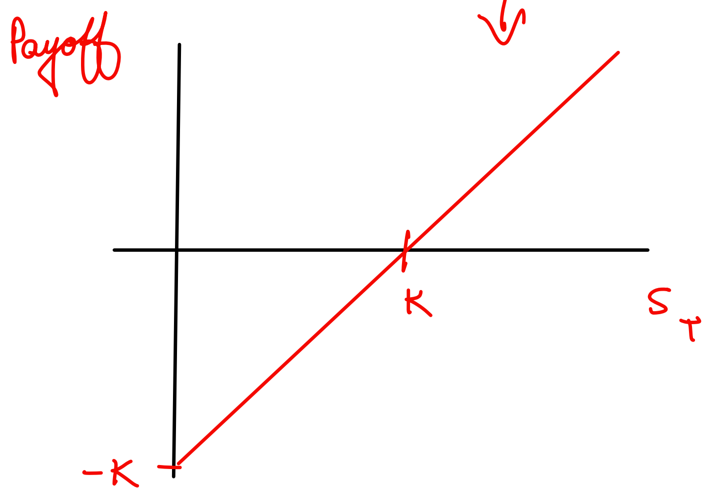
$4\frac{12}{23}$ ———————— $19\frac{1}{24}$

ONGC → Sell
RIL → Buy

(S_T) spot price → price on the spot (on that day)
(F_T) Futures price → a future price we will be seeing today, but will be paying/getting on expiry date

Arbitrage opportunity → Buy low, sell high

	Buy posn	Sell posn
Payoff	$S_T - K$	$K - S_T$



$K \to$ constant

The day you enter the contract

$F_T = K$

But on next day, maybe futures price (F_T) will change for same expiry date (for same contract)

We will be stuck with K

What is cash settlement?

↳ they will only exchange the difference in cash

↳ Not actually crude oil

Contract

Buyer → 6207 → Seller

On maturity date

Sell to market 6250 Buying from mkt & giving U/L to buyer

Mostly → dealt in cash only (instead of actually buying the commodity)

Futures contracts

↳ "Marked to market" daily

National Value $= \underbrace{6207}_{K \text{ per BBL}} \times \underbrace{100}_{\text{lot size}} \times \underbrace{50}_{\text{No. of contracts}}$

10% of NV → Initial margin

The moment we enter a futures contract, we have to maintain a margin with the exchange → A leverage

Exchanges do not do anything special other than forwards but, they give the option to exit by taking short position

15/1/24

Q On July 1, 2021 co enters into long contract to buy 10mn JPY in Jan 1, 2022
On Sep 1, 2021 it " " short " " sell " " " " Jan 1, 2022
Describe the payoff

Soln

Payoff

Long : $S_T - K$
Short : $K - S_T$

Buy	Sell	Mat date
July 1	Sep 1	Jan 1, 2022
F_1	F_2	$S_T - F_1$
		$F_2 - S_T$
		$F_2 - F_1$

Hedger

ONGC

RIL/HPCL

Bid → price at which exchange would buy from you
Ask → " " " " sell to you

Short ₹6267/BBL ₹6376 long/BBL

27/11/23 18/12/23

Loss in futures market = ₹(6267 − 6376) = ₹109/BBL

Sell ₹6376
− 109
₹6267

Exchange buys		Exchange sell
Bid		Ask
5775	5781	5787
Investor sells		Investor buys

speculators

Arbitrageur → profit without any risk (arbitrage)

Buy today : Gold S_0: ₹64,000/10gm

Short : F_{3m}: ₹65,000/10gm

r_f: 5% pa

On 15/04/24

Receive : 65k

Pay : $64\left(1+\frac{5}{400}\right) = 64.8k$

₹200 (Profit)

If there's such a arbitrage opportunity, everybody will take a short position

Q2 Trader buys two future contract on orange juice. Each contract is for delivery of 15,000 pounds. $F_0 = 160$ cent/lb. Initial margin is \$ 6000 per contract, maintenance margin is \$ 4500 per contract. What price change would lead to a margin call? Under what circumstance would \$ 2000 be withdrawn from margin account?

In futures contract, we don't pay the amount early, we just have to maintain the margin

Ans

$$\frac{15,000 \times 1.6 - (6000 - 4500)}{15,000} = 1.6 - 0.1 = 1.5 \$/lb$$

$$= 150 \text{ cent}/lb$$

price change $= 160 - 150 = 10$ cent/lb

Note:- You can't withdraw anything below \$ 6000 and whenever the margin goes below \$ 4500, you get a margin call

So, if my margin is suppose \$ 5500, then I won't be able to withdraw money, nor would I get a margin call.

Basis

gain $= F_1 - F_2$

Loss $= F_2 - F_1$

Payoff $= S_2 + F_1 - F_2$ → Net cash inflow to the seller

$= S_2 - (F_2 - F_1)$ → cash outflow to the buyer

Forward contract → can be settled at the date decided mutually by both parties

Futures " → settled on specific date (last Thursday of every month)

Basis $= b_t = S_t - F_t$

Payoff $= S_2 + F_1 - F_2$ → Net cash inflow to the seller $= F_1 + S_2 - F_2$

$= F_1 + \underbrace{b_2}$

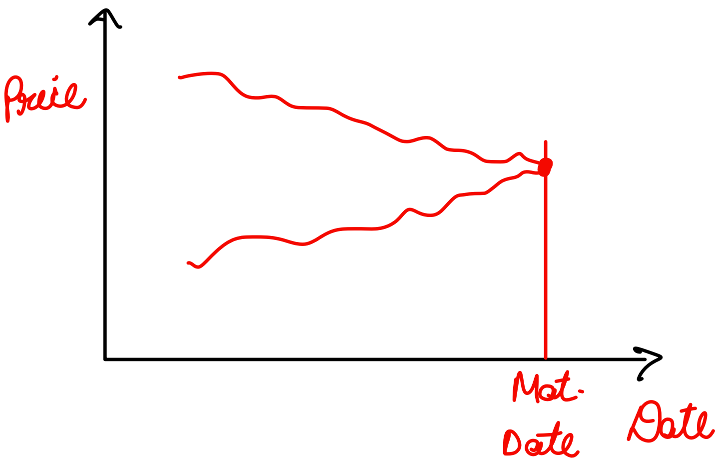
this I know, bcz on the day I am entering the market

this is the risky part

I am not sure about b_2

→ Basis can be +ve or -ve

→ Basis is the source of risk



Cross hedging → want to hedge risk of one underlying, but future not available, so we hedge using other asset and risk depends on two things

① Underlying asset

② Price diff. b/w the two assets

$$S_2 + F_1 - F_2 + S_2^* - S_2^*$$

$$\underset{USD}{F_1} + \underset{CNY}{(S_2 - S_2^*)} + \underset{USD}{(S_2^* - F_2)}$$

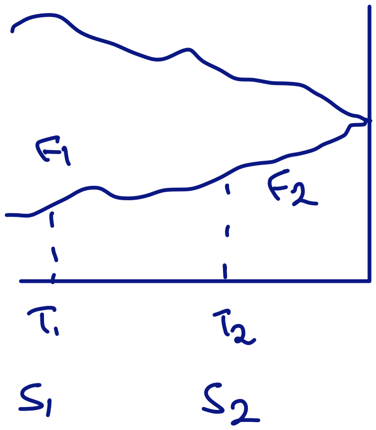
USD CNY USD USD USD

$S_2 \to$ exchange rate of chineese yuan

I want to take a short posn in CNY, but not available, so I take USD

17/1/24

Rolling over → Jan - Feb - March - April



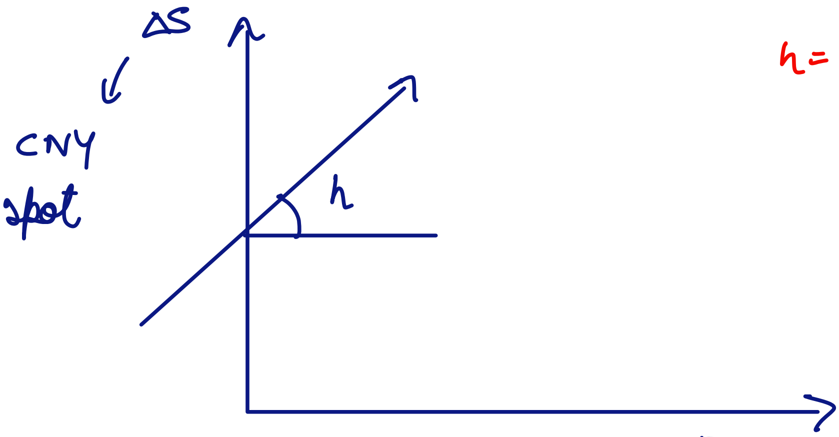
$S_2 + (F_1 - F_2)$

$F_1 + \underbrace{(S_2 - F_2)}_{b_2}$

I don't know S_2, F_2 which includes risk

If I wait till maturity S_2, F_2 converge and my risk becomes 0, as F_1 is certain on maturity

h = hedge ratio
↳ how many contracts I need to buy for hedging



$$h = \frac{\mathrm{cov}(\Delta F, \Delta S)}{\mathrm{Var}(\Delta F)} = \rho_{\Delta F, \Delta S} \cdot \frac{\sigma_{\Delta S}}{\sigma_{\Delta F}}$$

$$\# \text{ contracts} = h \frac{V_A}{V_F} = HR \times \frac{\text{Value of asset to be hedged}}{\text{Value of 1 future contract}}$$

$$= \frac{-0.01 \times 2 \times 10^7 \times 11.5523}{83.17 \times 1000} = -28.729$$

$$R_i = r_f + \beta(R_m - r_f)$$

$$R_i - r_f = \beta(R_m - r_f)$$

$$\beta = \frac{cov(R_m - r_f, R_i - r_f)}{Var(R_m - r_f)}$$

22/1/24

$P_0 = 47928.62$ Cr

$\beta = 1.2$

Short futures contract with U/L as Nifty 50

lot size = 50 $\quad$ ($F_0 = 21{,}765$)

$$\# \text{contract} = \beta \frac{P_0}{\text{Fut Value}}$$

$$= \frac{1.2 \times 47{,}928.62 \times 10^7}{21765 \times 50}$$

$$= 5{,}28{,}503$$

Dividend yield (annual) $= 4\%$

Time to maturity = 2 months $= \frac{2}{12}$

Dividend yield over 2 months $= 4\% \times \frac{2}{12}$

$$\text{Capital gain} = \frac{P_2 - P_1}{P_1}$$

$$\text{Dividend yield} = \frac{D_2}{P_1}$$

Total return = Cap gain + Div yield

Nifty 50 today = 21,571.80

2 month Nifty 50 = 21,000

Cap gain = -0.02651

$R_m = -0.01984$

r_f = 6% annually

$$E(R_i) = r_f + B(E(R_m) - r_f)$$

$$= 6\% \times \frac{2}{12} + 12\left(-1.98 - 6\% \times \frac{2}{12}\right)$$

$$= -2.58\%$$

Expected Value = 4.66917 E+11

gain in fut market $= \left[E(F_{2m}) - F_0\right] \times 50 \times \#$ contracts

3.16. The standard deviation of monthly changes in the spot price of live cattle is (in cents per pound) 1.2. The standard deviation of monthly changes in the futures price of live cattle for the closest contract is 1.4. The correlation between the futures price changes and the spot price changes is 0.7. It is now October 15. A beef producer is committed to purchasing 200,000 pounds of live cattle on November 15. The producer wants to use the December live cattle futures contracts to hedge its risk. Each contract is for the delivery of 40,000 pounds of cattle. What strategy should the beef producer follow?

3.30. It is July 16. A company has a portfolio of stocks worth \$100 million. The beta of the portfolio is 1.2. The company would like to use the December futures contract on a stock index to change the beta of the portfolio to 0.5 during the period July 16 to November 16. The index futures price is currently 1,000 and each contract is on \$250 times the index.
(a) What position should the company take?
(b) Suppose that the company changes its mind and decides to increase the beta of the portfolio from 1.2 to 1.5. What position in futures contracts should it take?

Assumptions

- Participants come for investment purpose
- r_f, borrowing = r_f, lending
- No transaction cost (No brokerage / tax)
- Buy / Sell fractional contract

Future price

No known income till T on U/L asset

$21{,}571 = S_0$

$21{,}765 = F_0 = S_0 e^{r_f T}$

If $F_0 > S_0 e^{r_f T}$

Sell (for F_0) / Buy (for $S_0 e^{r_f T}$)

borrowing at risk free rate

Borrow S_0 @ r_f $\rightarrow$ Return $S_0 e^{r_f T}$

Buy U/L

Enter into short contract @ F_0 $\rightarrow$ Recieve F_0 by selling U/L

$\Downarrow$

riskless profit

Soon everyone will do this, people will take short posn and soon it will be normal

If $F_0 < S_0 e^{r_f T}$

Buy low (for F_0) / Sell high (for $S_0 e^{r_f T}$)

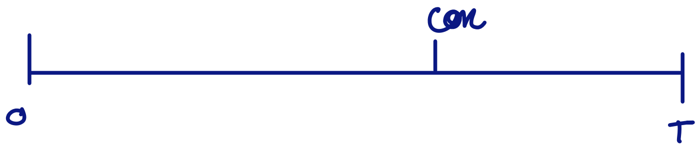
depositing at risk free rate

Short the asset @ S_0

Keep ₹S_0 @ r_f till T $\rightarrow$ Recieve $S_0 e^{r_f T}$

Enter into long contract @ F_0 $\rightarrow$ Buy U/L by paying F_0

So, people will look to short the asset, so $S_0 \downarrow$, $F_0 \uparrow$ till $S_0 e^{r_f T} = F_0$

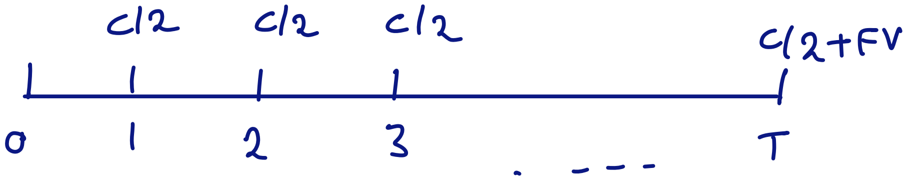


$$F_0 = \left[S_0 - PV(\text{Income})\right] e^{rT}$$

31|1|24

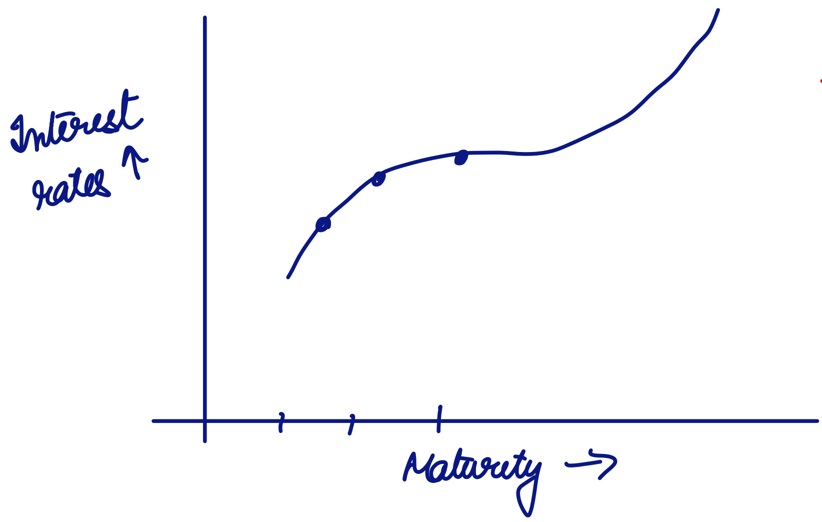
$$R_c = m \ln\left(1 + \frac{SAIR_m}{m}\right)$$

$$SAIR_m = m\left[e^{R_c/m} - 1\right]$$



6 months

$$P = \frac{c/2}{\left(1+\frac{r_1}{2}\right)} + \frac{c/2}{\left(1+\frac{r_2}{2}\right)^2} + \frac{c/2}{\left(1+\frac{r_3}{2}\right)^2} + \cdots \frac{c/2+FV}{\left(1+\frac{r_T}{2}\right)^T}$$

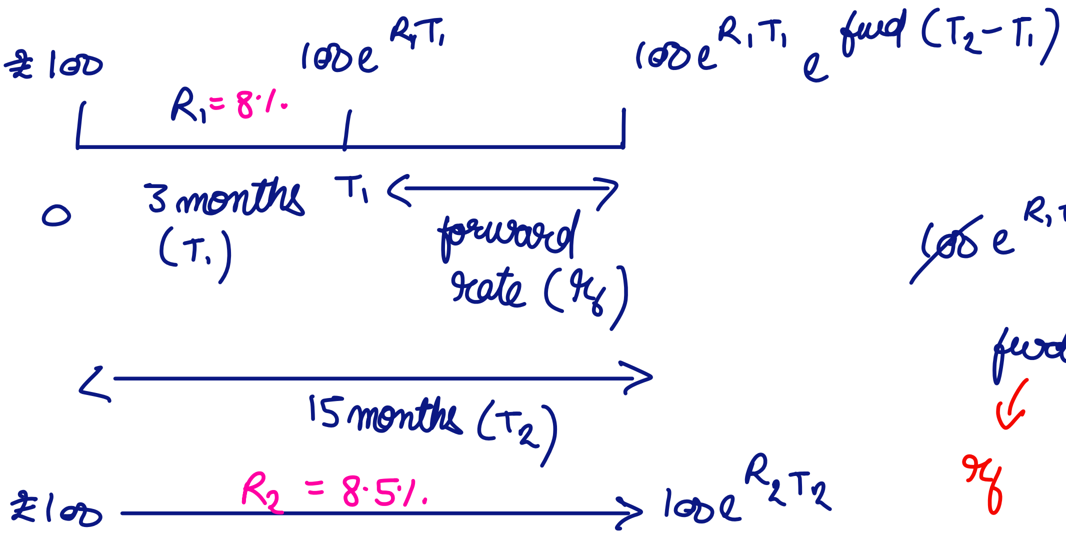


→ Zero coupon yield curve (ZCYC)

↓

for different maturities you have different interest rates

Forward rates



Using logic of arbitrage

$$\cancel{100}e^{R_1T_1}e^{fwd(T_2-T_1)} = \cancel{100}e^{R_2T_2}$$

$$fwd = \frac{R_2T_2 - R_1T_1}{T_2 - T_1}$$

↓

r_f

If actual / market interest rate > calculated forward rate

$$= \frac{8.5\% \times \frac{15}{12} - 8\% \times \frac{3}{12}}{\frac{15-3}{12}}$$

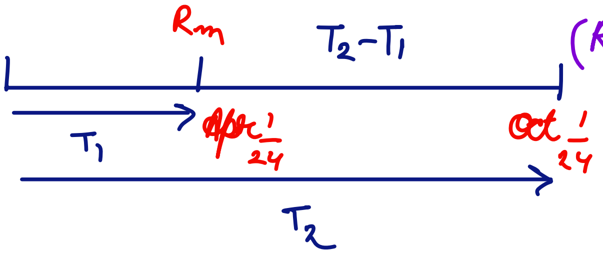
r_f → later converted to semi annual compounded interest rate, so we can use simply $P \times R \times T$

→ This forward rate calculation is based on Today's Term structure / interest rate. For Tomorrow it will again change and again calculated

5/2/24

Based on this we have a forward rate agreement
↓
Just like a forward, but on interest rates

→ what it could have received (X)



$$\underbrace{(R_K - R_m) \times L \times (T_2 - T_1)}_{\text{Notional payment X would be getting}}$$

X lends L in future
Y borrows L " "

R_K → decided rate (what X is receiving)
R_m → what X could have received

Interest rate $= R_K = 7\%$

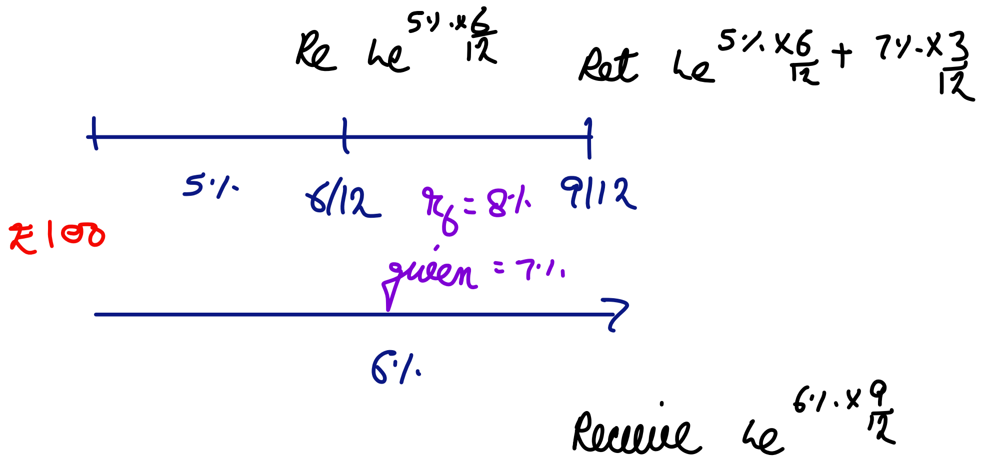
$$\text{Payoff} = \frac{(R_K - R_m) \times L \times (T_2 - T_1)}{1 + R_m(T_2 - T_1)}$$

L → notional principal
Nobody will actually lend or borrow principal, they will simply exchange interest rate differences

$$V_{FRA} = (R_K - r_f)\, L\, (T_2 - T_1)\, e^{-r_2 T_2}$$

↳ Value of forward rate agreement today

4.30. A bank can borrow or lend at LIBOR. Suppose that the 6-month rate is 5% and the 9-month rate is 6%. The rate that can be locked in for the period between 6 months and 9 months using an FRA is 7%. What arbitrage opportunities are open to the bank? All rates are continuously compounded.



So, arbitrage opportunity

Borrow L @ 5% for 6^m

Lend L @ 6% for 9^m

Enter into borrowing position in FRA for amount $Le^{5\% \times \frac{6}{12}}$

If market rates < Theoretical Rates

↳ then borrow in this rate

If market rate > Theoretical rates

↳ then lend in this rate

Q 18 4.75%

24 5%

value of FRA that promises to pay 6% (annual compounded)

notional principal = 10^6

for 6 month period starting in 18 months

$$r_f(cc) = \frac{5\% \times 24 - 4.75\% \times 18}{24-18} = 5.75\%$$

[continuous compounding]

$$r_f(sa) = e^{r_{cc} \times 1} = \left(1 + \frac{R_m}{m}\right)^m$$

$$R_m = m\left(e^{R_{cc}/m} - 1\right)$$

$$r_f(sa) = 2\left(e^{r_f/2} - 1\right) = 5.8335\%$$

r_R = 6% annually compounded

$r_R(sa)$ =

$$\left(1 + \frac{R_{sa}}{2}\right)^2 = 1 + R_a$$

$$R_{sa} = 2\left[(1+R_a)^{1/2} - 1\right]$$

$$= 5.9126\%$$

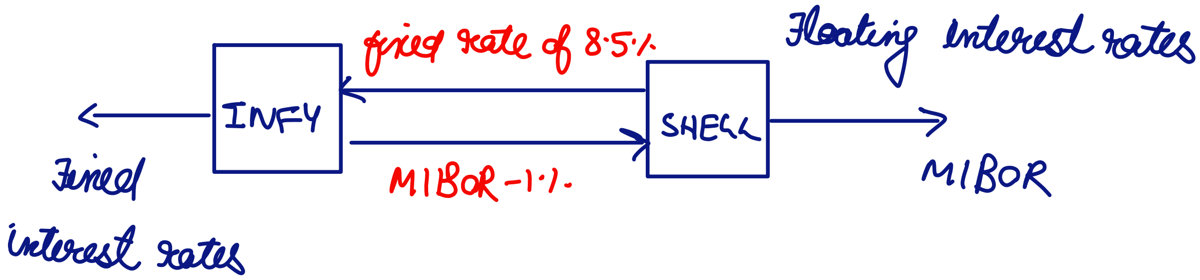
$$V_{FRA} = \frac{(5.9126 - 5.8335)}{100} \times 1mn \times \frac{6}{12} e^{-5\% \times \frac{24}{12}}$$

Use discrete compounding → for calculating payoffs

Use cont. " → for valuation

SWAPS → borrowing & lending to each other

Interest rate swaps



Infosys → had liability in fixed interest rate, changed it to floating rate

Pays int @ 8%
" " @ MIBOR - 1%
Recv. " @ 8.5%

Net pays [8% + MIBOR - 1% - 8.5%]

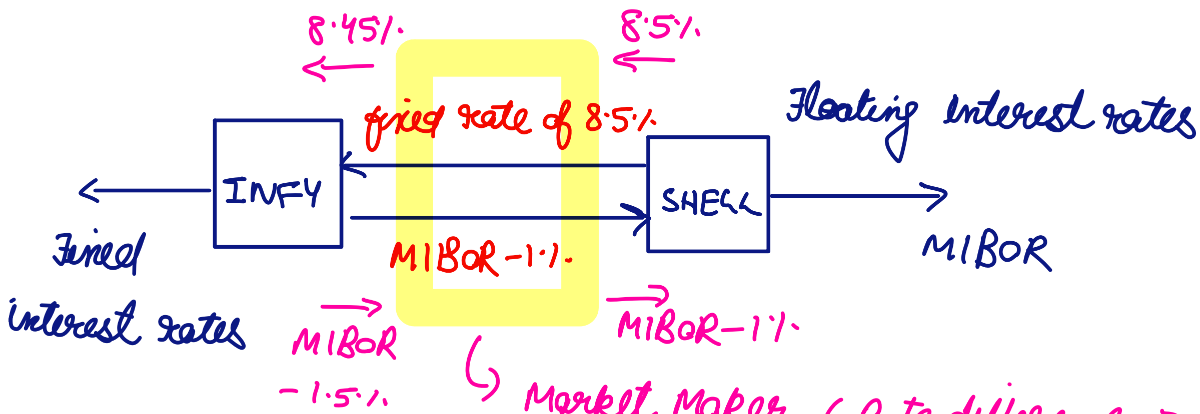
= MIBOR - 1.5%

Shell

MIBOR + 8.5% - (MIBOR - 1%)

9.5%

Market Maker → Bank



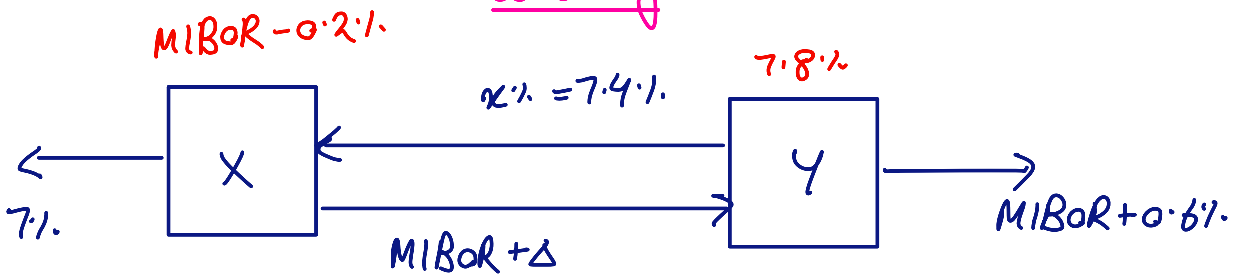
Market Maker (Rate difference → earnings)

In this case liability is considered, similar thing can be applied for assets (like interest being received)

Comparative Advantage

	Fixed	Floating
X : AAA	7%	MIBOR
Y : BB	8%	MIBOR + 0.6%
	1%	0.6%

Borrowing



$$X:\ 7\% + MIBOR + \Delta - x\% \le MIBOR$$

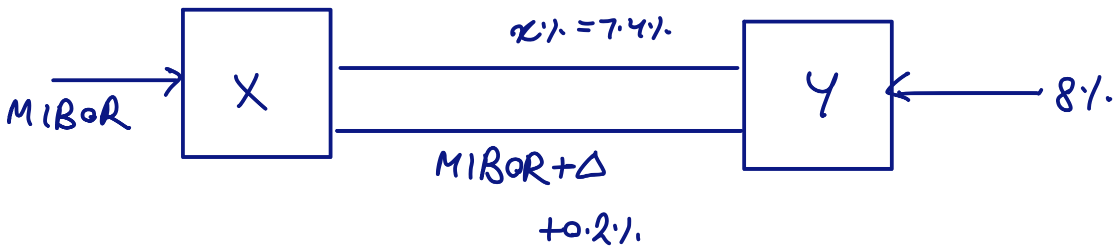
$$7\% \le x - \Delta$$

$$\cancel{MIBOR} + 0.6\% + x\% - \cancel{MIBOR} - \Delta \le 8\%$$

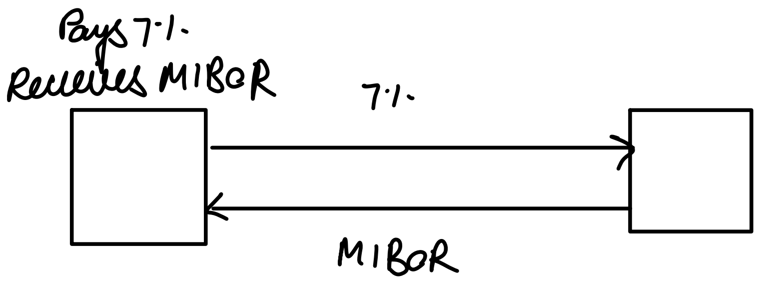
$$x - \Delta \le 7.4\%$$

So, swap helped them reduce interest rates

Lending



7/2/24



$L = 100$

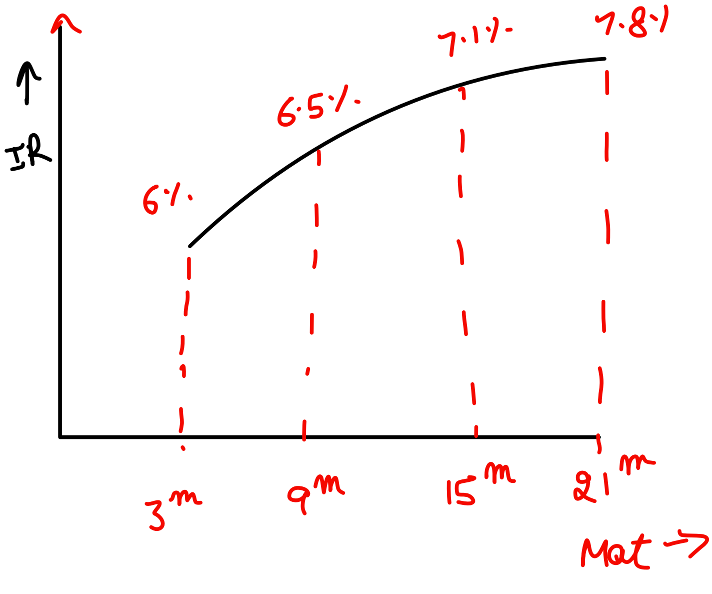
The MIBOR on last payment date will be used for next payment

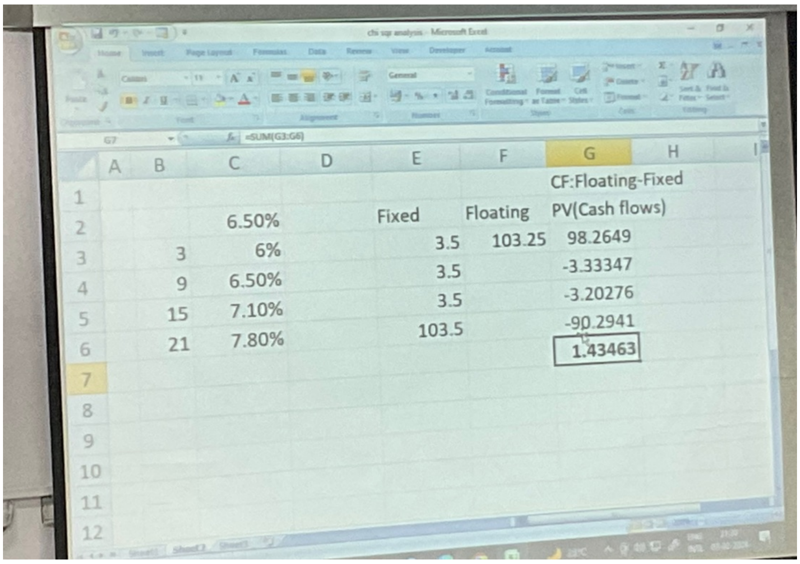
-3^m	0	3^m	9^m	15^m	21^m	
		3.5	3.5	3.5	100+3.5	Payments
		3.25	3.4325	4.081	4.891	Received
		100				

Last interest payment day

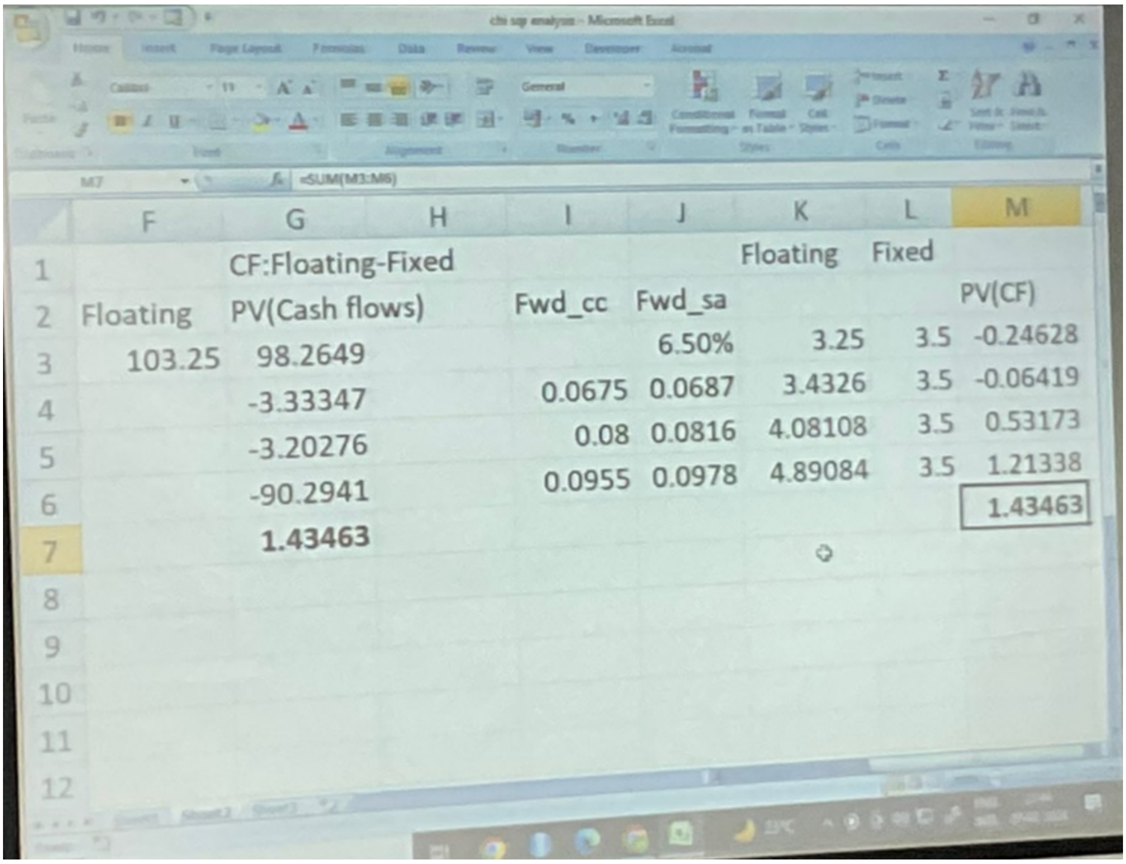
	r_1	r_2	r_3	r_4
	C	C	C	FV+C

$$P_0 = FV = \frac{C}{1+r_1} + \frac{C}{(1+r_2)^2} + \frac{C}{(1+r_3)^3} + \frac{C+FV}{(1+r_4)^4}$$



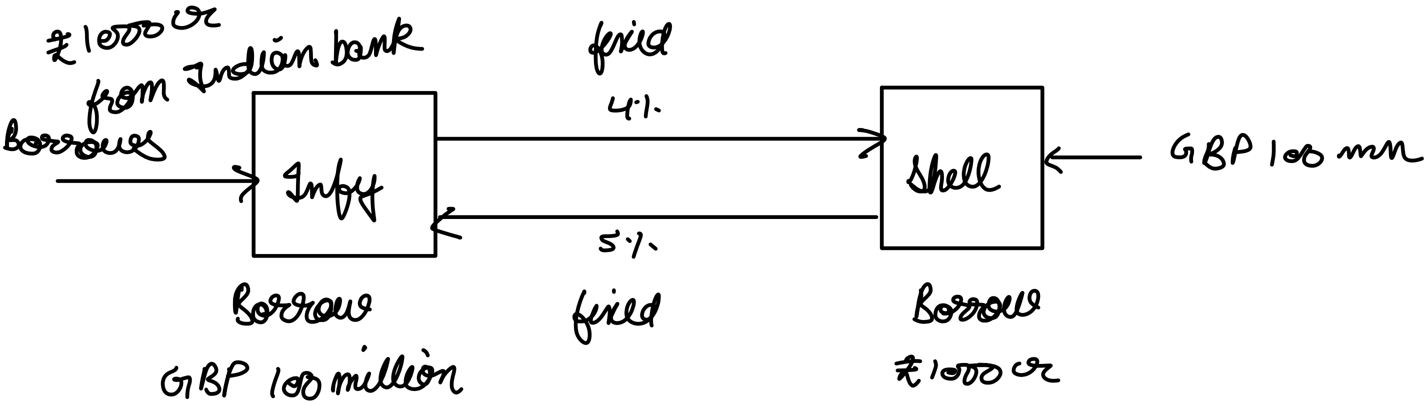


	B	C	E	F	G
1					CF:Floating-Fixed
2		6.50%	Fixed	Floating	PV(Cash flows)
3	3	6%	3.5	103.25	98.2649
4	9	6.50%	3.5		-3.33347
5	15	7.10%	3.5		-3.20276
6	21	7.80%	103.5		-90.2941
7					1.43463



M7 =SUM(M3:M6)

	F	G	H	I	J	K	L	M
1		CF:Floating-Fixed				Floating	Fixed	
2	Floating	PV(Cash flows)		Fwd_cc	Fwd_sa			PV(CF)
3	103.25	98.2649			6.50%	3.25	3.5	-0.24628
4		-3.33347		0.0675	0.0687	3.4326	3.5	-0.06419
5		-3.20276		0.08	0.0816	4.08108	3.5	0.53173
6		-90.2941		0.0955	0.0978	4.89084	3.5	1.21338
7		1.43463						1.43463
8								
9								
10								
11								
12								



fixed to fixed currency swap

Post Midsem

28|2

options

Call option - option to buy an underlying asset

↳ buying the option to buy U/L asset

↳ but the seller of the call option is obliged to sell

Put option → option to sell an U/L asset

↳ but the buyer of put option is obliged to buy.

American option → can exercise at any pt

Europen " → exercise at maturity

In the money, at the money, out of the money

stock split → for every 2 stocks, they will make it into 5, 3 stocks

Ex 5 for 2 [2 stocks splitted into 5]

$2 \times P_0 = 5 \times P_1$ P_0 → original price

$\frac{2}{5} P_0 = P_1$ P_1 → after stock split

So, K will be revised → $\frac{2}{5} K$

buy 1 stock @ ₹100

↓

buy $\frac{5}{2}$ stock @ ₹40

$\underbrace{\text{Bonus shares, stock split}}_{\text{same effect}}$

underlying price & no. of underlyings are revised

but for cash dividends this is not revised max $(S_T - K, 0)$

European $\left\{ \begin{array}{cc} c & C \\ p & P \end{array} \right\}$ American

max $(K - S_T, 0)$

Call → spend K on T

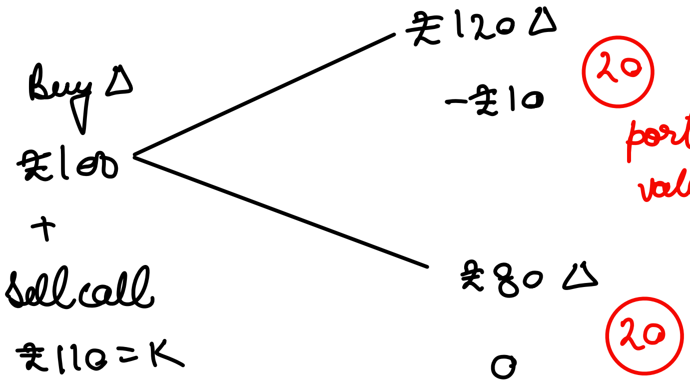
Today Ke^{-rT} [hold less cash]

Put → recv. K on T

	c	p
Underlying price ↑	↑	↓
K ↑	↓	↑
Time to maturity ↑	↑	↑
Return Volatility of U/L asset ↑	↑	↑
r_f ↑	↑ ↓ ?	↑ ↓ ?

Add Notes!!

Binomial Trees

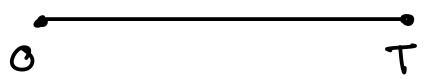


$120\Delta - 10 = 80$

$\Delta = \frac{10}{40} = 0.25$

$100\Delta - c = 20e^{-r_f T}$

$c = 25 - 20e^{-r_f T}$

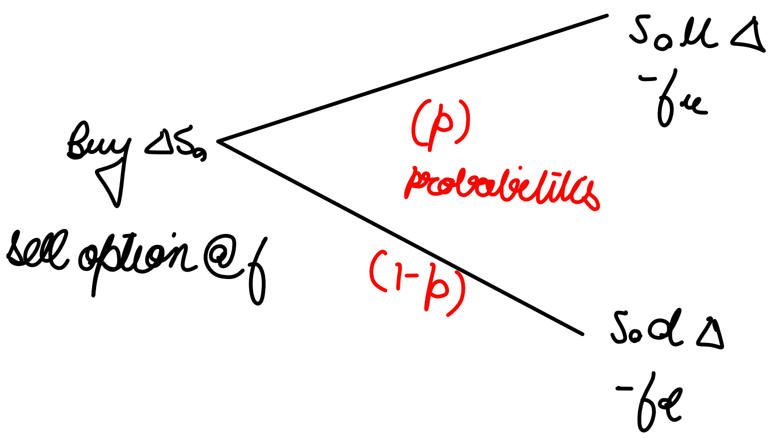


Riskiness of the stock is in 120 & 80 after that you have decided the riskiness, it gets converted to risk free and we can discount using risk free rate

We are not moving out of riskiness of stock.

$u = 1.2$

$d = 0.8$



$S_0 u\Delta - f_u = S_0 d\Delta - f_d$

↳ If we can get such a Δ, we can have a risk free

$$\Delta = \frac{f_u - f_d}{S_0(u-d)}$$

$$\Delta S_0 - f = \underbrace{(S_0 u\Delta - f_u)}_{\text{can be } S_0 d\Delta - f_d} e^{-r_f T}$$

$$f = \Delta S_0 - (S_0 u\Delta - f)e^{-r_f T}$$

$$f = \frac{f_u - f_d}{u-d} - \left(\frac{u f_u - u f_d}{u-d} - f_u\right) e^{-r_f T}$$

$$= \frac{(f_u - f_d) - (d f_u - u f_d)\, e^{-r_f T}}{u-d}$$

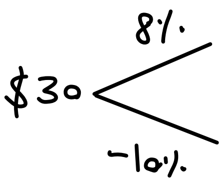
$$= \frac{f_u\,(1 - d e^{-r_f T})}{u-d} + \frac{f_d\,(u e^{-r_f T} - 1)}{u-d}$$

$$= e^{-r_f T}\left[f_u \frac{e^{r_f T} - d}{u-d} + f_d \frac{u - e^{r_f T}}{u-d} \right]$$

$$f = e^{-r_f T}\underbrace{\left[p f_u + (1-p) f_d \right]}_{\text{like an expected value}} \quad \text{where } p = \frac{e^{r_f T} - d}{u-d}$$

$u, d \to$ includes the risk, we still don't know u & d

Q u/l



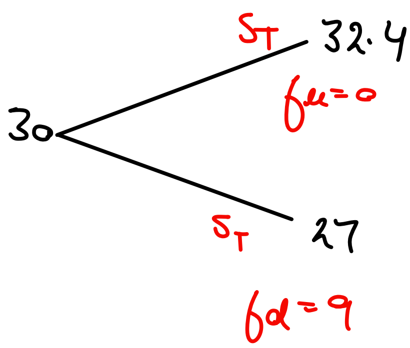
$u = 1.08$

$d = 0.9$

$r_f = 5\%$

$T = 4$ months

payoff $= \max(30 - S_T, 0)^2$



$$32.4\,\Delta = 27\Delta - 9$$

$$\Delta = \frac{-9}{5.4}$$

$$-32.4 \times \frac{9}{5.4} = -54, \quad -\left(27 \times \frac{9}{5.4} + 9\right) = -54$$

$$30\Delta - f = -54\, e^{-5\% \times \frac{4}{12}} = 53.1075$$

$f = -50 + 53.1075 = 3.1075$

$\Delta = \dfrac{f_u - f_d}{u - d} = \dfrac{0 - 9}{30(1.08 - 0.9)} = \dfrac{-9}{5.4}$

$p = \dfrac{1.017 - 0.9}{1.08 - 0.9} = 0.65$

$f = e^{-rT}[0.65 \times 0 + 0.35 \times 9] = e^{-5\% \times 12} \times 0.35 \times 9 = 3.1075$

11/4/24

stochastic process ↳ process through which random variable gets its values

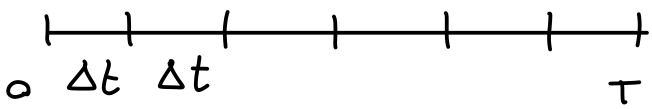
discrete — continuous

Markov Process → Y → RV following markov process
↳ past prices not important
only value today is sufficent to predict the future

Y ~ Markov process → current values relevant to predict future

weak form of efficiency → No technical analysis

Markov Process — Weiner Process



$Z_0 \quad Z_{0+\Delta t}$

$\Delta Z = \varepsilon\sqrt{\Delta t}$

$N\Delta t = T$ [N divisions]

$\Delta z \rightarrow$ in any time period is independent of each other

$\varepsilon \sim \phi(0,1)$

$\Delta Z = \varepsilon\sqrt{\Delta t}$

$\Delta t \rightarrow$ are equal

$\Delta z \rightarrow$ not necessarily equal, and are independent

$\Delta z \sim N(0, \Delta t)$

$z(T) - z(0) = \underbrace{\sum_{i=1}^{N} \Delta z_i}$ → adding normal distributions

$\sim N(0, N\Delta t)$ → Independent so no correlation

$Z(T) - Z(0) \sim N(0, T)$

Q $Z_0 = 100$

$\Delta Z_{1y} \sim N(0,1)$

$\Delta Z_{3m} \sim N(0, \frac{1}{4})$

what is probability that $\Delta Z_{3m} <$ some value → Question type

generalised WP

$\Delta x = a\Delta t + b\Delta Z$ → $Z \sim WP$

$= a\Delta t + b\sqrt{\Delta t}\,\varepsilon$ → here is the uncertainty

$\Delta x \sim N(a\Delta t, b^2\Delta t)$

deterministic

stochastic part

→ Here a, b constants

If $\Delta x = a\Delta t$

$dx = a\,dt$ $\quad \Delta t \rightarrow 0$

$x = x_0 + at$

Ito's Process → special generalised WP

$$\Delta x = \underbrace{a(x,t)}_{\text{Values at today}} \Delta t + \underbrace{b(x,t)}_{\text{Values at today}} \Delta Z$$

→ Here $a, b \to f(x,t)$

$$\Delta x \sim N\left(a(x,t)\Delta t,\ b^2(x,t)\Delta t\right)$$